Ruixiang (Kevin) Wu

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Education

University of Illinois, Urbana-Champaign Sept 2023 – May 2027
BS in Computer Science and Mathematics

- GPA: 3.92/4.0
- **Undergrad Coursework:** Algorithm and Models of Computation, Nonlinear Programming, Real Analysis, Probability Theory, Data Structure, Applied Linear Algebra, Prob & Stat for Computer Sci
- **Graduate Coursework:** Convex Optimization, Intro to Optimization, Iterative Method, Parallel Numerical Algorithm

Research Interests

Optimization, Machine Learning, Numerical Linear Algebra, Algorithm
My research focuses on low-rank optimization, large-scale numerical methods and preconditioner, with an emphasis on designing efficient algorithms and understanding the geometry of modern learning problems.

Research Experience

Simultaneous Lyapunov Stability: Accelerating SDP Feasibility Detection via Preconditioners May 2025 – present
Advisor: Prof. Richard Y. Zhang and Prof. Madeleine Udell

- Nyström Sketch Preconditioner
  - Naive GD
  - Trust-region method
  - Conjugate Gradient
  - Riemannian GD
  - Truncated CG
  - Exact Proximal method
  - Iterior-Point method
- Riemannian Tangent-space Preconditioner
  - Gauss-Newton
- Primal-dual
  - Dual-hessian preconditioner
  - Adaptive preconditioner

Parallel OLS with Randomly Pivoted Cholesky for Sparse Approximation Jan 2026 – present
Advisor: Prof. Edgar Solomonik

- Reformulated sparse approximation via Orthogonal Least Squares as pivoted partial Cholesky on the Gram matrix and proved $O(1/k)$ convergence bounds for greedy and randomized pivoting.
- Developed a block randomized OLS algorithm with rejection sampling to support parallel pivot selection while preserving the correct sequential sampling law.
- Tested the method on ill-conditioned and clustered dictionary matrices, showing that rejection sampling matches sequential randomized OLS and avoids failures of naive block sampling.

Global Guarantees for Unconstrained and Nonnegative Low-rank Matrix Recovery Jan 2025 – July 2025
Advisor: Prof. Richard Y. Zhang

- Study theoretical guarantees for nonnegative matrix factorization and low-rank matrix factorization.
- Establish global landscape results showing the absence of spurious local minima under the model studied.
- Construct descent and negative-curvature directions to certify strict-saddle behavior at non-optimal stationary points.

Projects

Efficiency of Coordinate Descent Methods on Huge-Scale Optimization Problems

- Read and synthesize research literature on coordinate descent methods for large-scale optimization.
- Summarize convergence guarantees and practical implementation considerations.
- Analyze when coordinate descent is preferable to full-gradient methods in high-dimensional settings.

Talks

UIUC SIAM student seminar presentation March 2026

Presented work on preconditioned large-scale optimization.

Teaching Experience

Course Assistant for CS 225 Data Structure Jan 2024 – May 2024

- Assisted with weekly lab sections.
- Held regular office hours and supported students with course material and programming assignments.

Awards

Dean's list Spring 2024 - Spring 2026

Technical Skills

Languages: Matlab, Python, LaTeX

Technical Areas: Optimization, Mathematical Modeling, Simulation, Numerical Linear Algebra, Algorithms.